The Most Beautiful Equation in Finance:

Integrating Cash Capital and Risk Capital to Value Anything

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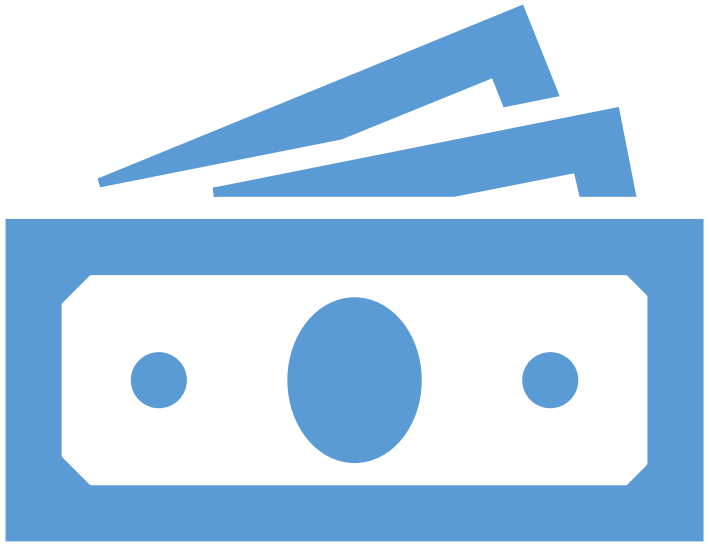
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The Motivation:

A proof by counter-example

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How do you value a risky cash flow in the CAPM/APT?

- A risky asset produces a positive expected terminal cash flow in one year
  - Mean $\mu > 0$, Variance σ^2
  - CAPM $r_i = r + \beta_i(r_m - r) = r + \pi$
    - r is the risk-free rate
    - π is the risk premium
- CAPM/APT valuation
  - $V_0 = \frac{\mu}{1+r+\pi} > 0$

What if the cash flow risk increases?

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“What would Elon Musk do?”

- The offer: On the toss of a fair coin, sometime now or in the future, Elon wins or loses $10B
  - Expected value = 0
- What is the deal value?
- CAPM/APT says zero, regardless of risk adjustment
- **However, Elon Musk should *absolutely refuse* the deal, because he has more productive ways to take risk.**
- He has the capacity to take risk, even though the deal involves no cash
- **His risk capacity (not cash) is a valuable asset.**
- **He must earn a return on his risk capacity**



Note: This is not a risk aversion argument.

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Is there something wrong with the formula?

- $V_0 = \frac{\mu}{1+r+\pi} \rightarrow V_0 = \frac{\mu-\pi V_0}{1+r} = \frac{mean - risk\ charge}{1+r}$

Implicit assumption!

- *However, the risk charge πV_0 is assumed proportional to the asset's value*

- But that is how it has always been!
  - Markowitz – every asset has a **percentage** return and standard deviation
  - Sharpe – the risk premium is determined to be a **percentage** of an asset's value
  - Harrison/Kreps/Pliska – First Fundamental Thm of Asset Pricing **assumes** positive prices
  - Cochrane (and others) – **Assumes** positive SDF (stochastic discount factor)

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“What would Jamie Dimon do?”

- He’ll take the deal if you pay JPM
- Risk charge (RAROC) 25% of max loss
  - Max loss is $10B
- Stand-alone deal valuation is **–$2.5B** or maybe even lower due to concentration risk
- Valuation formula:
- $V_0 = (\mu - Risk\ charge)B$
  - B is a risk-free discount factor



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Alternative Theory

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How to fix the textbook models

- Recall $V_0 = \frac{\mu - \pi V_0}{1+r}$

- **Generalize πV_0--- replace by $k\sigma$**
  - k is the cost of risk, a dollarized Sharpe Ratio
  - σ is a dollar measure of risk, not necessarily standard deviation

- Let $\mu = E[V_1 + C_1]$, the expected value of the asset at time 1 plus cash flow

- Then $V_0 = \frac{\mu - \pi V_0}{1+r} \rightarrow V_0 = \frac{\mu - k\sigma}{1+r} \rightarrow$

- $rV_0 + k\sigma = E[V_1 - V_0] + E[C_1]$

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The highest bid meets the lowest offer…

- Buyer expects returns to meet or exceed requirements
- Seller expects future returns to equal or fall short of requirements
- How can they both agree to trade?

Seller $\geq$

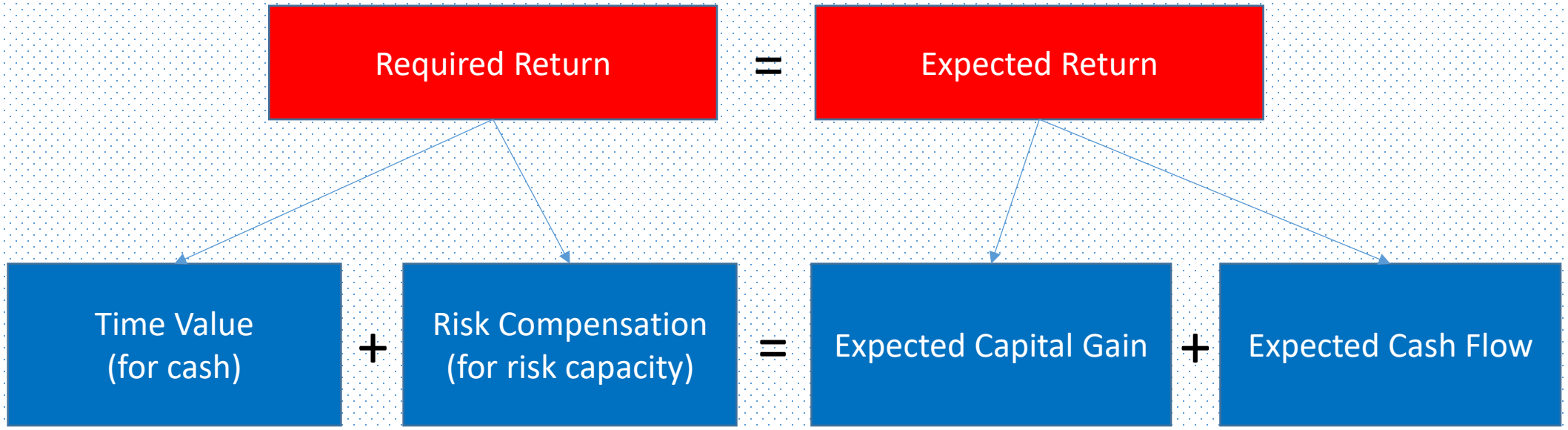
Required Return	=	Expected Return

$\leq$ Buyer

If the buyer and seller have equal information and requirements, they can only trade when these two returns are equal.

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Required and expected returns each have two components



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Returns should always be measured in **dollars**, *not percent* …

Trade	Meaningful percentage return?	Meaningful dollar return?
Buy securities	Yes ☑	Yes ☑
Buy securities on margin	No	Yes ☑
Short sell securities	No	Yes ☑
Long or short futures/forwards	No	Yes ☑
Long options	Yes ☑	Yes ☑
Short options	No	Yes ☑

… in order to evaluate all possible trades.
Sharpe Ratios should be dollarized.

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Time value

(opportunity cost of cash)

V_0 = asset value today

r = constant risk-free periodic discount rate

Time value = interest earned = rV_0

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Risk compensation

σ = risk measure per period, not necessarily a standard deviation

k = compensation per unit of risk, comparable to a Sharpe Ratio

$k\sigma$ = risk compensation per period

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Cash flow

$E[C_1]$ = Expected cash flow in one period

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Capital gain

expected change in price

V_0 = Value today

$= PV\{C_1, C_2, C_3, \ldots\}$

V_1 = Value just after C_1 is paid

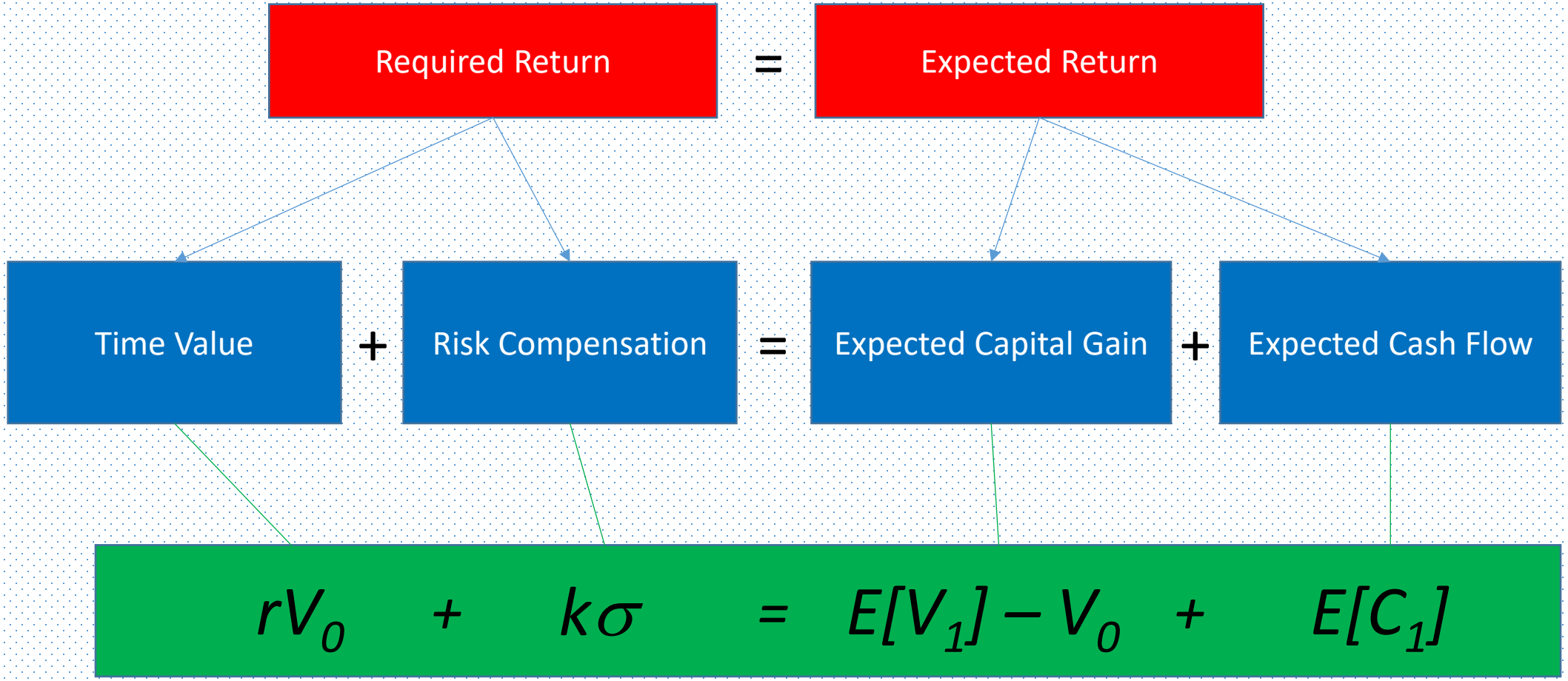
$= PV\{C_2, C_3, C_4, \ldots\}$

Expected capital gain

$= E[V_1] - V_0$

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Now put them together



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This is the most beautiful equation in finance

- The GVE, or **General Valuation Equation**

Time Value	+	Risk Compensation	=	Expected Capital Gain	+	Expected Cash Flow

$$rV_0 + k\sigma = E[V_1] - V_0 + E[C_1]$$

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Does the GVE work with…

Yes!

- Complex risk-free cash flow sequences?
- Models of derivative pricing?
- Risk-neutral pricing?
- Stochastic cash flows?
- Correlated stochastic cash flows over time?
- Portfolios of cash flows?
- Stochastic interest rates?



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Simple Pleasures

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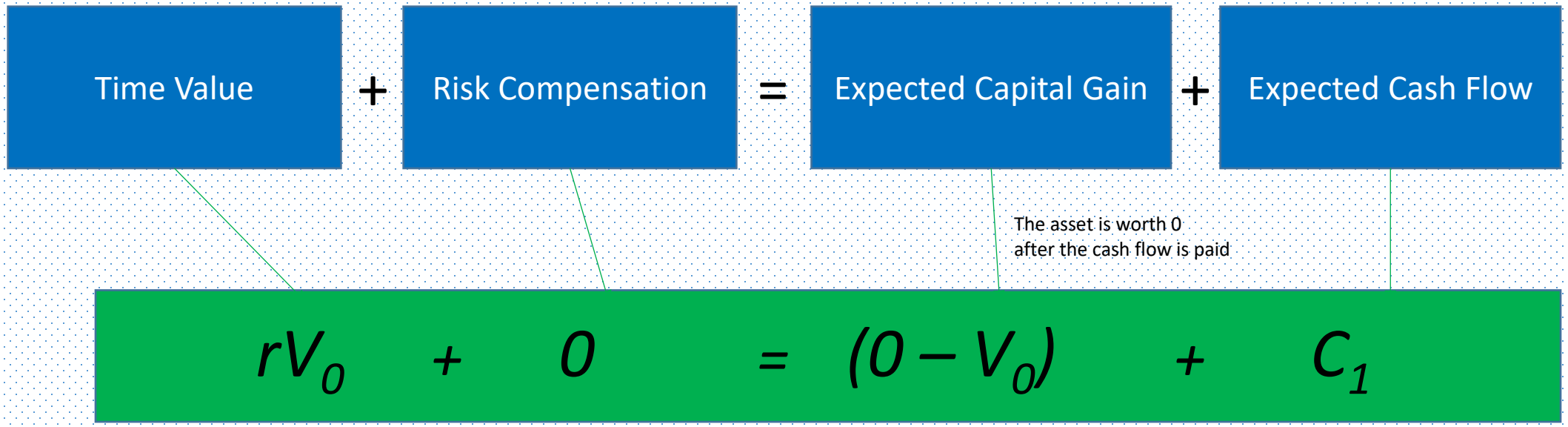
Let's value some risk-free cash flows

- Single cash flow
- Flat perpetuity
- Exponential growth perpetuity
- Flat annuity
- Loan amortization
- Exponential growth annuity
- Linear growth perpetuity
- Quadratic growth perpetuity
- Fibonacci growth perpetuity

No risk charges in this section.

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A guaranteed payment of C_1 in one period



GVE #1

$$V_0 = \frac{C_1}{(1+r)}$$

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Remember when you learned to value a flat perpetuity?

The traditional method notices that because of value-additivity of risk-free cash flows, we already have an equation for its valuation:

$$V_0 = \sum_{i=1}^{\infty} \frac{C}{(1+r)^i}$$

Note that the constant C has replaced the variable C_i. To simplify this, let $d = 1/(1+r)$ for convenience. Then we have

$$V_0 = Cd \times [1 + d + d^2 + d^3 + \ldots]$$

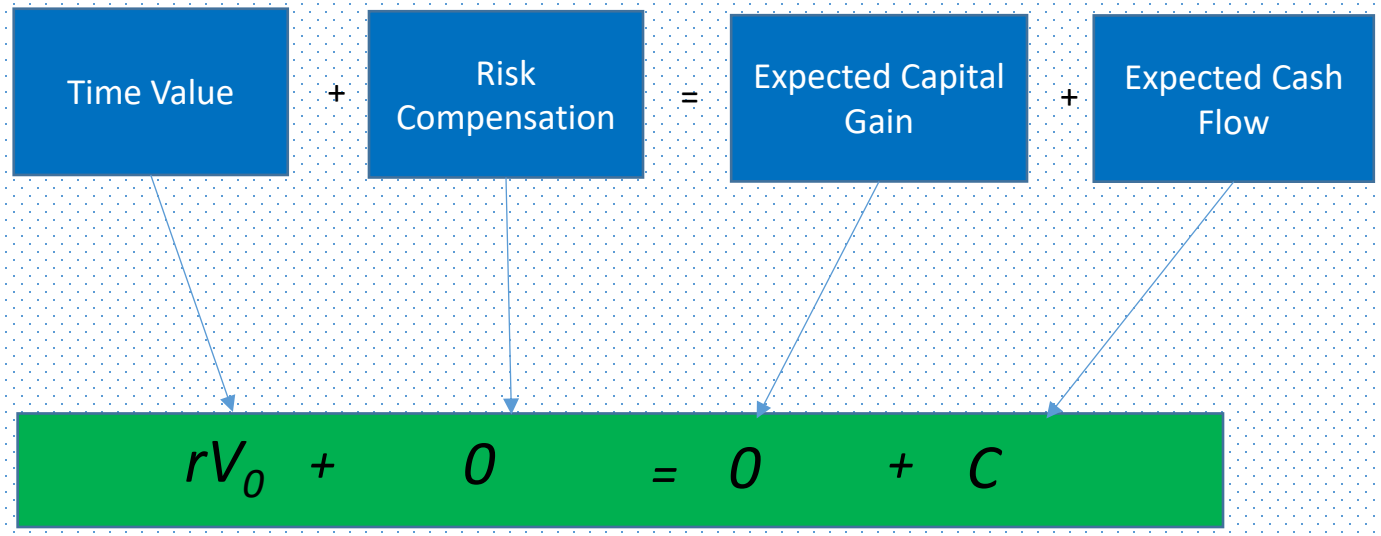
The term in brackets is an infinite geometric series, which converges as long as d<1, i.e. r>0. Therefore

$$[1 + d + d^2 + d^3 + \ldots] = 1/(1-d)$$

$$V_0 = Cd/(1-d) = C/r \quad \text{for } r>0$$

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A guaranteed flat perpetuity using the GVE



GVE #2

$$V_0 = \frac{C}{r}$$

Notes:

The cash profile {…} is the set of payments starting in one period

Let V_0 = PV{C, C, C, …}

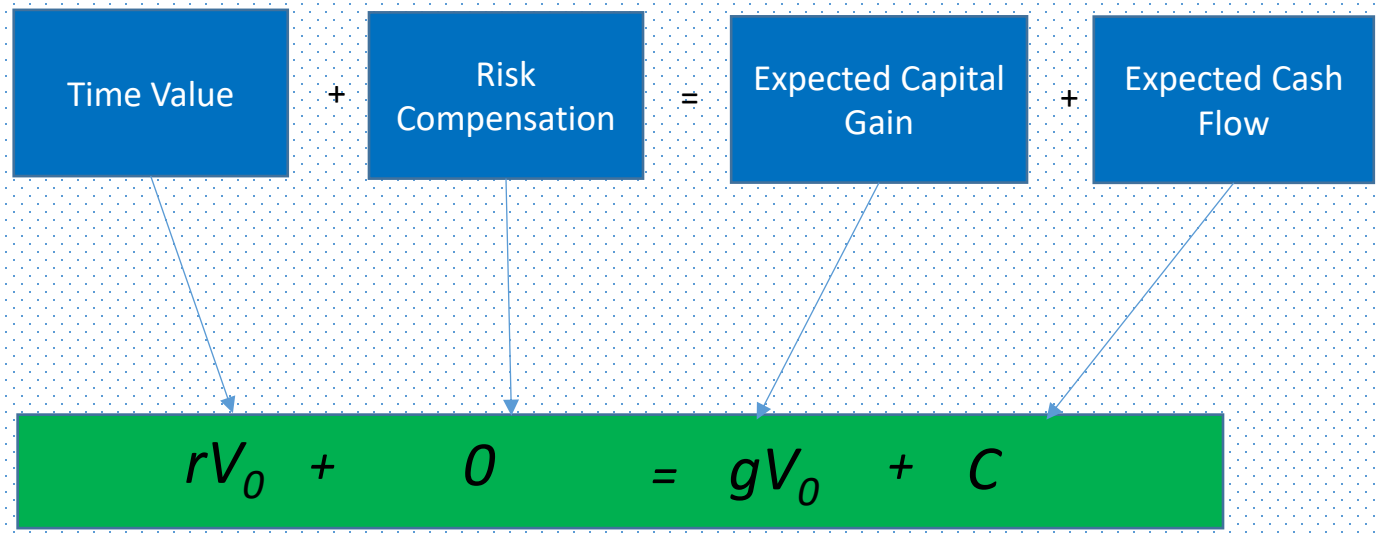
Then V_1 = PV{C, C, C, …}

And $V_1 - V_0$ = PV{0, 0, 0, …} = 0

This is the capital gain.

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An exponential growth perpetuity



GVE #3

Converges only if r > g

Notes:

The cash profile {…} is the set of payments starting in one period

Let $V_0 = PV\{C, C(1+g), C(1+g)^2, \ldots\}$

Then $V_1 = PV\{C(1+g), C(1+g)^2, C(1+g)^3 \ldots\}$

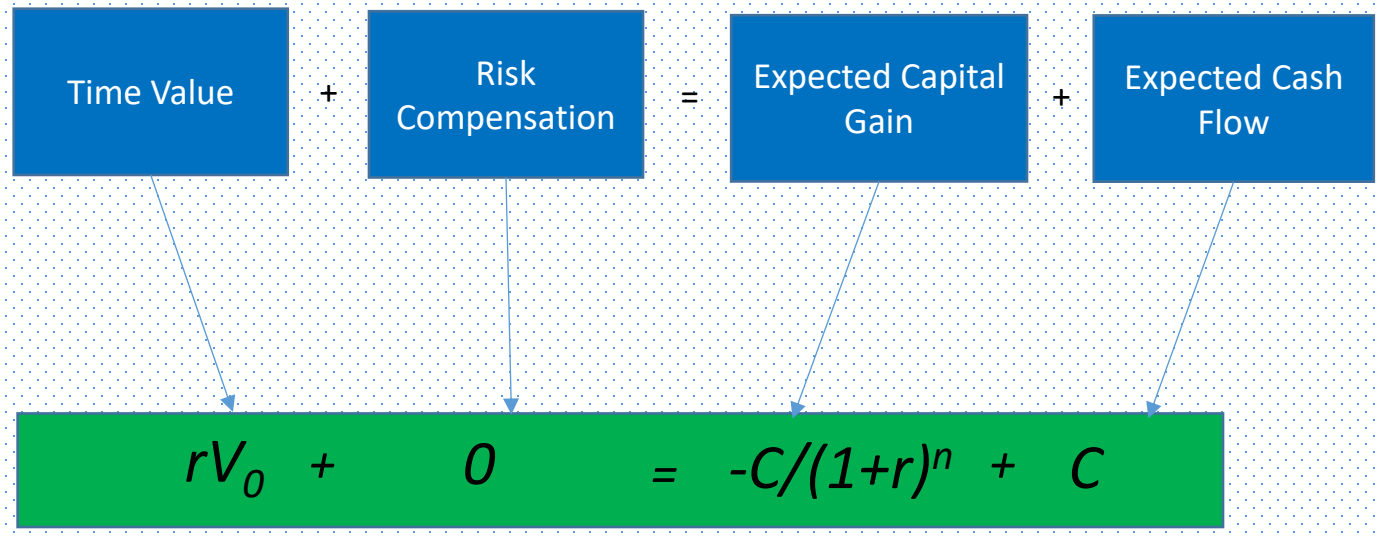
Or $V_1 = (1+g)\, PV\{C, C(1+g), C(1+g)^2, \ldots\}$

So $V_1 = (1+g)V_0$

$V_1 - V_0 = gV_0$

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A flat annuity



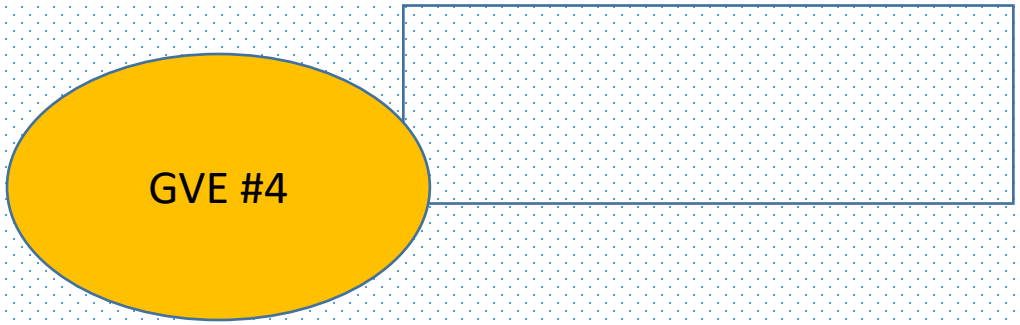
Notes:

Let $V_0 = PV\{C, C, C, \ldots C, C\}$ for n payments

Then $V_1 = PV\{C, C, C, \ldots C, \mathbf{0}\}$ for n payments

$V_1 - V_0 = PV\{0, 0, 0, \ldots, 0, -C)$
with $-C$ in the nth position

$V_1 - V_0 = -C/(1+r)^n$



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Finance 101

End = Begin + Interest – Payment

$$V_1 = V_0 + rV_0 - C$$

$$rV_0 = V_1 - V_0 + C$$

This is, of course, the GVE.

Loan Amortization 12 months

Loan amount	1000	
Interest rate	0.50%	monthly
Payment	$86.07	

Period	Begin	Interest	Payment	Principal Reduction	End
1	1000.00	5.00	86.07	81.07	918.93
2	918.93	4.59	86.07	81.47	837.46
3	837.46	4.19	86.07	81.88	755.58
4	755.58	3.78	86.07	82.29	673.29
5	673.29	3.37	86.07	82.70	590.59
6	590.59	2.95	86.07	83.11	507.48
7	507.48	2.54	86.07	83.53	423.95
8	423.95	2.12	86.07	83.95	340.01
9	340.01	1.70	86.07	84.37	255.64
10	255.64	1.28	86.07	84.79	170.85
11	170.85	0.85	86.07	85.21	85.64
12	85.64	0.43	86.07	85.64	0.00

V_0 rV_0 C $V_1 - V_0$ V_1

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An exponentially growing annuity!

Time Value + Risk Compensation = Expected Capital Gain + Expected Cash Flow

$$rV_0 + 0 = gV_0 - (1+g)C_0(1+g)^n/(1+r)^n + C_0(1+g)$$

$$r^*(1+g)V_0 + 0 = -(1+g)C_0/(1+r^*)^n + C_0(1+g)$$

GVE #5

$$V_0 = \frac{C_0}{r^*}\left(1 - \frac{1}{(1+r^*)^n}\right)$$

Converges even if r < g

Notes:

Let C_0 = time zero reference (unpaid)

$C_1 = C_0(1+g)$, $C_2 = C_0(1+g)^2$, …

Let $V_0 = PV\{C_1, C_2, C_3, \ldots C_n\}$

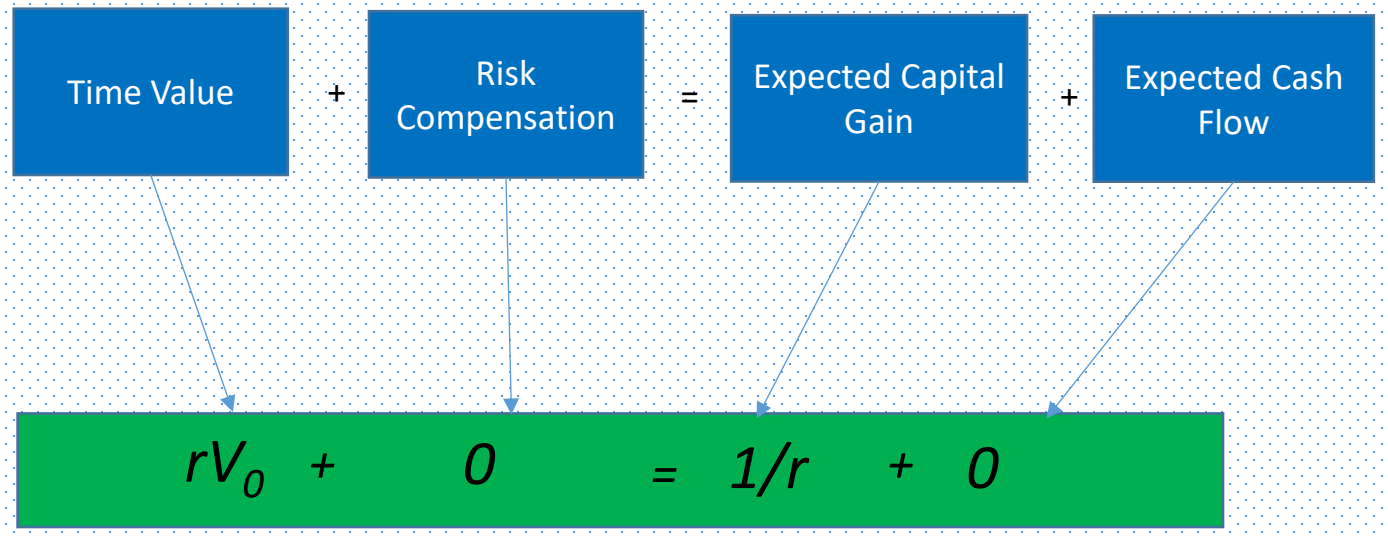
Then $V_1 = PV\{C_2, C_3, \ldots C_n, 0\}$ for n payments

$V_1 - V_0 = gV_0 - (1+g)C_0(1+g)^n/(1+r)^n$

Let $r^* = (r-g)/(1+g)$, $C_0 = C_1/(1+g)$

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A growing *linear* perpetuity pays {0, 1, 2, 3, …}



Notes:

Let $V_0 = PV\{0, 1, 2, 3, \ldots\}$

Then $V_1 = PV\{1, 2, 3, 4, \ldots\}$

And $V_1 - V_0 = PV\{1, 1, 1, \ldots\} = 1/r$

using GVE #5

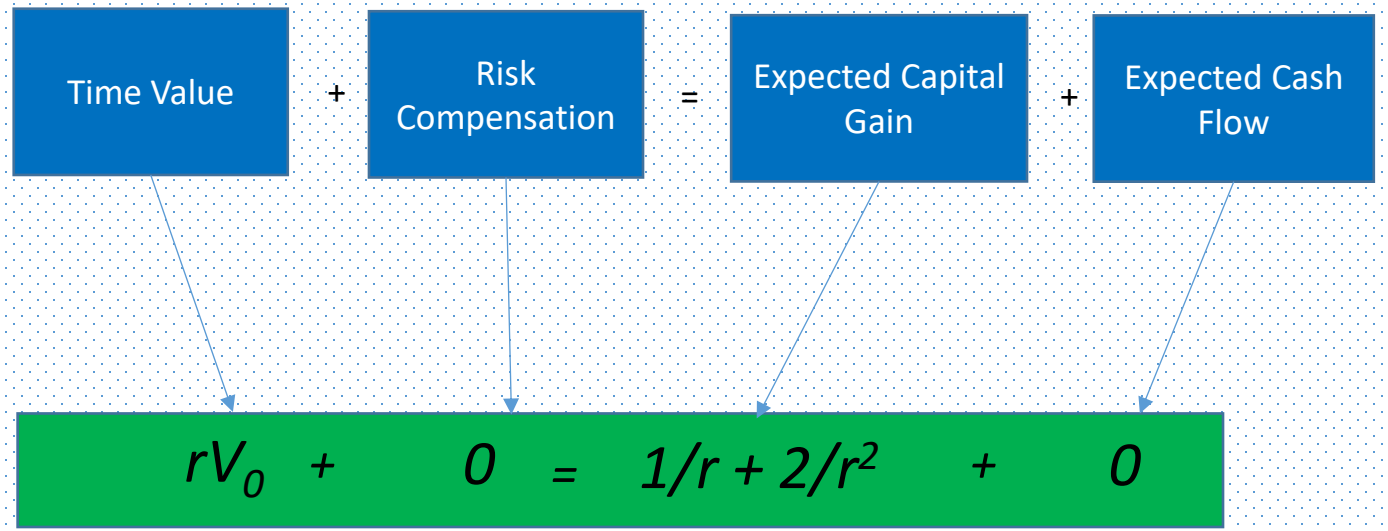
Have you ever seen this before?

GVE #6

Use this to compute the formula for duration

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A growing *quadratic* perpetuity pays {0, 1, 4, 9, 16, …}



Notes:

Let $V_0 = PV\{0, 1, 4, 9, \ldots\}$

Then $V_1 = PV\{1, 4, 9, 16, \ldots\}$

And $V_1 - V_0 = PV\{1, 3, 5, 7, \ldots\}$

$= PV\{1, 1, 1, \ldots\} + 2\ PV\{0, 1, 2, 3, \ldots\}$

$= 1/r + 2/r^2$

Using GVE #5 and #6

Or this?

GVE #7

Use this to compute the formula for convexity

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Note on duration and convexity

- Let e^{-ri} be the risk-free discount factor for i periods
- Then we have the bond's present value as a function of coupon and principal, assuming n periods to maturity
- $V_o = C\,\Sigma e^{-ri} + Pe^{-rn}$
- Take the derivatives with respect to V_0
  - First derivative
    - Duration = Rate of price change per unit discount rate change
    - $\frac{\partial V_0}{\partial r} = -C\,\Sigma i e^{-ri} - nPe^{-rn}$ → (negative) Linear growth annuity
  - Second derivative
    - Convexity = Rate of change of duration per unit discount rate change
    - $\frac{\partial^2 V_0}{\partial r^2} = C\,\Sigma i^2 e^{-ri} + n^2 Pe^{-rn}$ → Quadratic growth annuity

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More on continuously discounted bond prices

- Let $e^{-r(T-t)}$ be the risk-free discount factor for a (T-t) period interval
  - T = expiration date (principal paid), t = now
- Let the coupon payment C be made continuously, i.e. C *dt*
- Then the present value is an integral
- $V_t = \int_t^T Ce^{-rs}\,ds + P\,e^{-r(T-t)} = C\left(1 - e^{-r(T-t)}\right) + Pe^{-r(T-t)}$
- $rV_t\,dt = dt\left[rC\left(1 - e^{-r(T-t)}\right) + rPe^{-r(T-t)}\right] = "time\ value"$
- $dV_t = dt\left[-rCe^{-r(T-t)} + rPe^{-r(T-t)}\right] = \quad "capital\ gain"$
- $rV_t\,dt - dV_t = dt[\qquad\qquad\qquad

The Fibonacci sequence pays {0, 1, 1, 2, 3, 5, 8, …}

Time Value + Risk Compensation = Expected Capital Gain + Expected Cash Flow

$$rV_0 + 0 = (1+V_0)/(1+r) + 0$$

Notes:

Let $V_0 = PV\{0, 1, 1, 2, 3, 5, \ldots\}$

Then $V_1 = PV\{1, 1, 2, 3, 5, 8, \ldots\}$

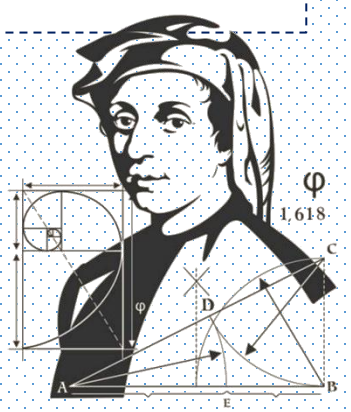
And $V_1 - V_0 = PV\{1, 0, 1, 1, 2, 3, \ldots\}$

This is the deferred Fibonacci series!

$V_1 - V_0 = 1/(1+r) + V_0/(1+r)$

GVE #8

$$V_0 = \frac{1}{r^2 + r - 1} = \frac{1}{(r - \phi + 1)(r + \phi)}$$

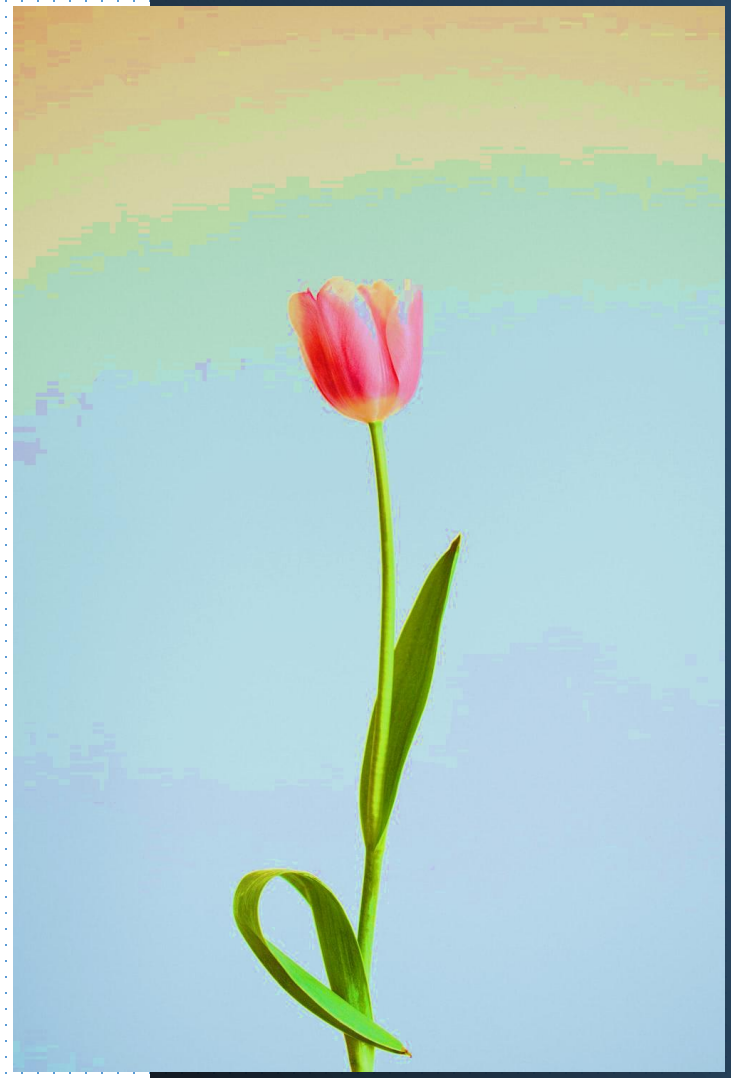


Infinite sequence converges only if 1 + r > 1.618, the Golden Ratio (ϕ)
The PV of the finite sequence can be computed for any r
as the difference between two growing annuities, using partial fractions.

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Why is the GVE so beautiful?

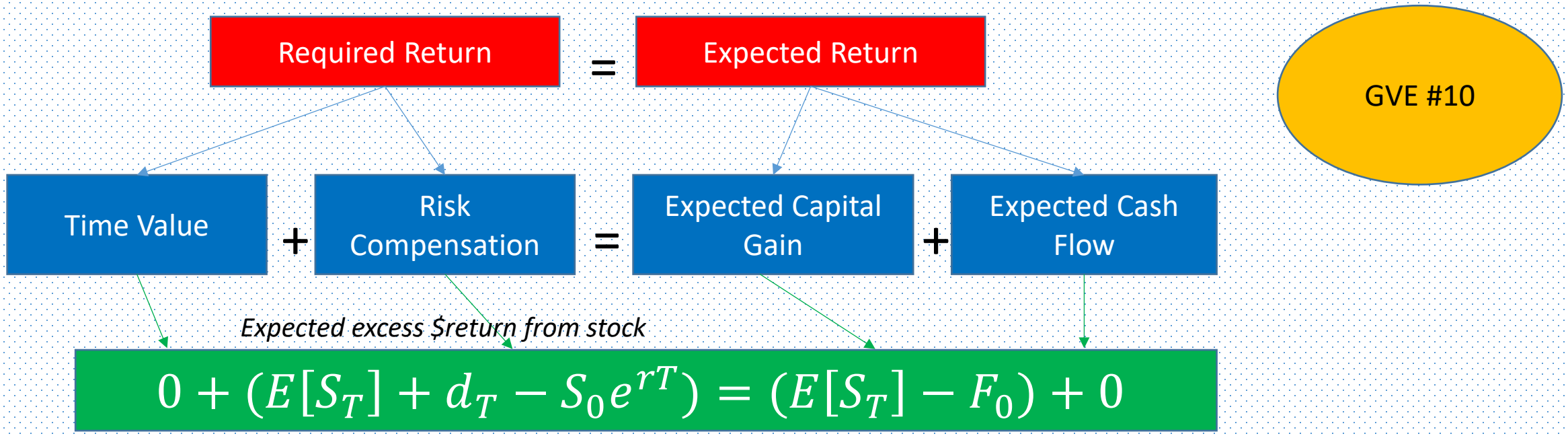
- It values many common cash flow patterns (with constant discount rate)
- It values assets in public and private portfolios
- It reconciles pricing differences between finance valuation models and financial institution valuation models
- It values derivatives
- It values risky cash flows with intertemporal correlations
- It can incorporate stochastic interest rates



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Financial forward contract –

Discrete time, continuous compounding, known dividends lost to long position



**dollar return notation*

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Forward Contract Notes

Notation – S_T is the underlying ex-dividend financial asset value at time T

Time value – The value of a forward contract at time 0, with no counterparty risk, $V_0 = 0$

Risk charge – The long forward position takes the same price risk as
the one who holds the underlying asset but does not receive dividends

Capital gain – The expected value of the forward position in the future

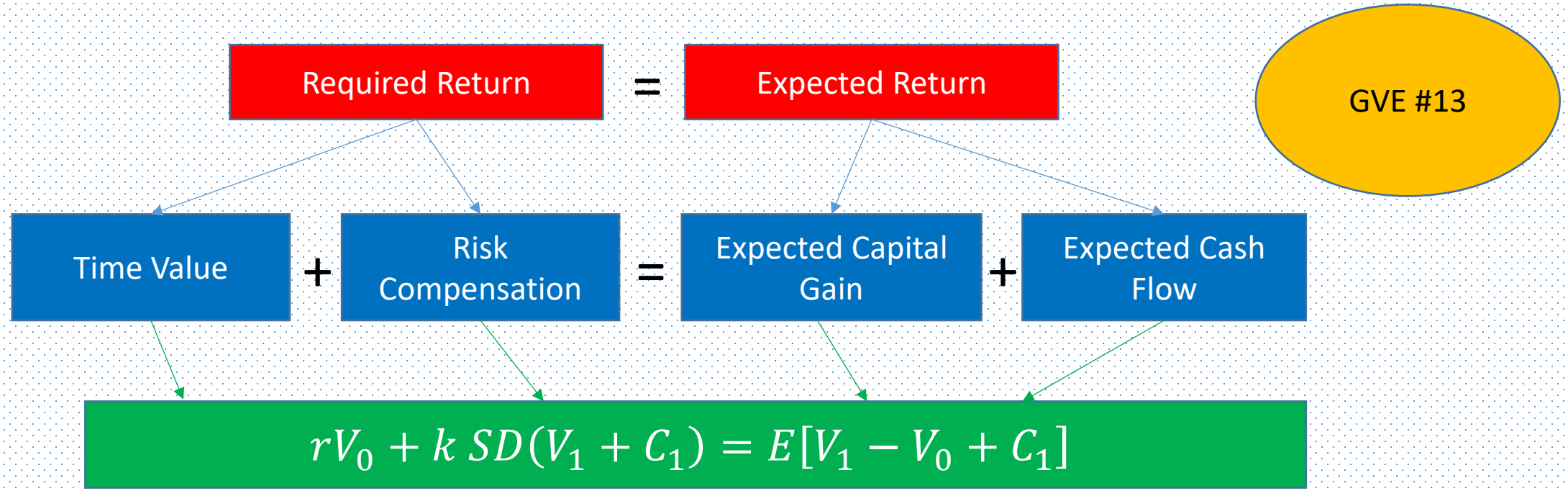
Cash flow – Assumed dividends are known; d_T is the future value of known dividends, compounded continuously at the risk-free rate

NOTE HOW THE $E[S_T]$ TERMS CANCEL –
EXPECTED FUTURE VALUE INCLUDES RISK COMPENSATION BUT ALSO AFFECTS EXPECTED RETURNS
THESE TWO THINGS CANCEL, LEADING TO A RISK-NEUTRAL VALUATION

$$0 + (E[S_T] + d_T - S_0 e^{rT}) = (E[S_T] - F_0) + 0$$

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Single risky cash flow $C_1 \sim N(\mu, \sigma)$ in one period



Once called certainty-equivalent (μ*) valuation;
Note limit as σ→∞ and compare to CAPM
Note obvious application to risk-neutral option pricing

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Rebuilding the theory of finance, Part 1

Returns must be measured in dollars in order to analyze all financial transactions. Never in percent.

The GVE allows both discrete-time and continuous-time formulations.

Utility functions may be replaced by specifying the risk measure and the cost of risk. These can be measured empirically, unlike utility.

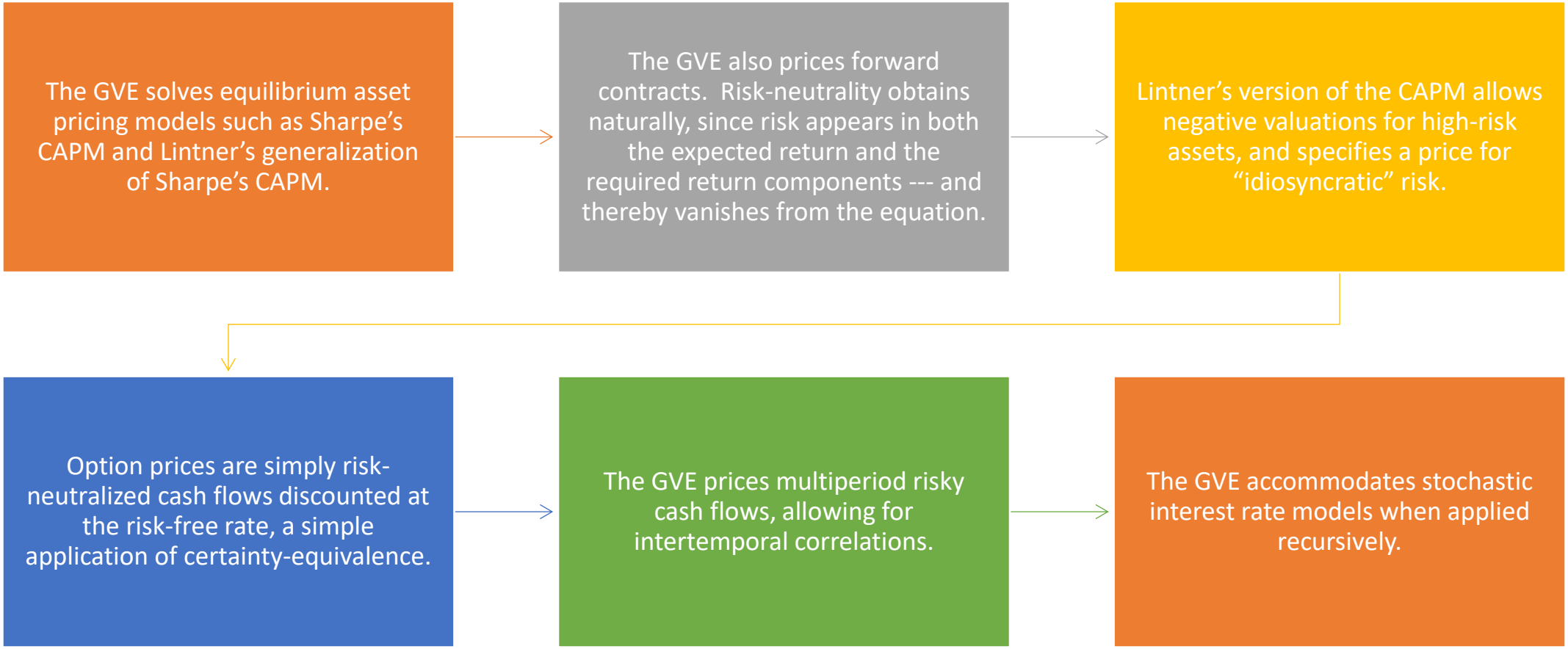
The GVE can be written as a difference equation or differential equation used to value any asset or contract.

Many risk-free cash flow sequences can be valued easily and intuitively with the GVE.

We should re-examine the practice of discounting at risk-adjusted rates. This is a special case of the GVE that leads to unsupportable valuations at high risk levels.

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Rebuilding the theory of finance, Part 2



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Convergence

$$V_0 = \frac{\mu - k\sigma}{1 + r}$$

General case

Valuation models with risk and capital charges used by **banks and insurance companies**

$$V_0 = \frac{\mu - \pi V_0}{1 + r}$$

$$V_0 = \frac{\mu}{1 + r + \pi}$$

Special case

Valuation models used in **corporate finance**

- Risk charge is assumed proportional to asset value
- Leads to adjusting the discount rate for risk
- Special case of the GVE
- Supportable only for well-capitalized assets because of calibration benefits of comparables

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Valuation models for risky cash flows should have these…

- Market-measurable parameters
- Separate compensation for cash capital and risk capital
- Negative valuations as risk grows large
- The ability to evaluate all assets, contracts, portfolios and trades: Including underfunded positions, futures, short options
- A means to determine the equilibrium risk and return parameters for optimal portfolio construction, including all assets
- A way to value correlated cash flows over time
- An allocation of risk charges over time
- The textbook models as a special case

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Let’s do valuation *the right way!*

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